SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA0325	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	K. Venkata Karthik	**Reg. No.:**	192372092
Section / Batch:	BE-CSE(AI)	**Date:**	12-10-2026

Code of Conduct

I K. Venkata Karthik / 192372092 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Venkata Karthik

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

Q1. A field engineer downloads a 15 MB firmware file on a mobile link that sustains an average of 12 Mbps. How long will the download take?

Ans: Size = 15 MB × 8 = 120 Mb; time = 120 / 12 = 10 s.

Q2. A background sync service on a smartphone draws an average current of 160 mA from a 4000 mAh battery. How many hours can the phone run this service before the battery is empty?

Ans: Time = 4000 mAh / 160 mA = 25 hours.

Q3. A handset sends a 1500-byte packet over a 2 Mbps uplink, and the propagation delay to the base station is 5 ms. What is the total one-way delay for the packet?

Ans: Transmission = 12000 bits / 2 Mbps = 6 ms; total = 6 + 5 = 11 ms.

Q4. A mobile app compresses a 4 MB photo by 75% before uploading it over a 400 kbps packet-data link. How long does the upload take?

Ans: After 75% compression: 1 MB = 8 Mb; time = 8,000,000 / 400,000 = 20 s.

Q5. A planner locates co-channel cells by moving i = 2 cells along one axis and then j = 1 cell after a 60° turn. What cluster size N does this produce?

Ans: $N = i^2 + ij + j^2 = 4 + 2 + 1 = 7$.

Q6. An operator adopts a 12-cell reuse cluster. What is the co-channel reuse ratio Q = D/R for this design?

Ans: $Q = \sqrt{(3N)} = \sqrt{36} = 6$.

Q7. Cells in a 7-cell cluster have a radius of 2 km. What is the minimum distance between the centres of two co-channel cells?

Ans: $D = R\sqrt{(3N)} = 2 \times \sqrt{21} = 2 \times 4.583 \approx 9.17$ km.

Q8. A district of 1000 sq. km is to be covered with hexagonal cells of 1 km radius. Approximately how many cells are needed?

Ans: Hexagon area = $2.598R^2 = 2.598$ km²; cells = 1000 / 2.598 ≈ 385 cells.

Q9. Each cell must support 50 full-duplex channels, each using 25 kHz in each direction, with a 7-cell reuse pattern. How much total spectrum does the operator need?

Ans: Channels per cluster = 50 × 7 = 350; bandwidth = 350 × (25 + 25) kHz = 17.5 MHz.

Q10. A town is served by 10 cells. Traffic growth forces every cell to be split into microcells of half the original radius. How many cells will cover the same area after splitting?

Ans: Halving the radius makes each cell ¼ the area, so 10 × 4 = 40 cells.

Q11. A cell with 60 channels is converted to three 120° sectors, with channels shared equally. How many channels does each sector get?

Ans: 60 / 3 = 20 channels per sector.

Q12. A system has 1000 duplex channels and adopts a 4-cell cluster. How many channels does each cell receive?

Ans: 1000 / 4 = 250 channels per cell.

Q13. A 7-cell reuse system operates where the path-loss exponent is 4. Considering only the six first-tier co-channel interferers, estimate the signal-to-interference ratio in dB.

Ans: S/I = (√(3N))^n / i0 = (√21)^4 / 6 = 441 / 6 = 73.5 ≈ 18.7 dB.

Q14. A designer tests whether a 4-cell cluster can meet an 18 dB S/I target with a path-loss exponent of 4 and six first-tier interferers. What S/I does N = 4 actually give, and is the target met?

Ans: Q = √12 = 3.464; S/I = Q^4 / 6 = 144 / 6 = 24 ≈ 13.8 dB. This is below 18 dB, so the target is not met.

Q15. A drive test records a serving signal of -70 dBm, co-channel interference of -88 dBm and a noise floor of -100 dBm. Compute the SINR in dB.

Ans: Interference + noise = 1.585×10^{-12} mW + 1×10^{-13} mW = 1.685×10^{-12} mW ≈ −87.7 dBm; SINR = −70 − (−87.7) ≈ 17.7 dB.

Q16. A 1800 MHz carrier is used for a GSM rollout. What is its wavelength?

Ans: λ = c / f = 3×10^{8} / 1.8×10^{9} ≈ 0.167 m (16.7 cm).

Q17. At 900 MHz, at what distance does the free-space path loss reach 120 dB?

Ans: FSPL = 32.45 + 20 log d(km) + 20 log f(MHz); 120 = 32.45 + 20 log d + 59.08, so 20 log d = 28.47 and d ≈ 26.5 km.

Q18. The received power at a 1 km reference distance is -50 dBm in an urban area with a path-loss exponent of 3.5. What power is expected at 4 km?

Ans: P = −50 − 10 × 3.5 × log(4/1) = −50 − 21.07 ≈ −71.1 dBm.

Q19. A 2100 MHz receiver can tolerate a maximum Doppler shift of 150 Hz. What is the highest vehicle speed, in km/h, the link can support?

Ans: λ = 3×10^{8} / 2.1×10^{9} = 0.143 m; v = fd × λ = 150 × 0.143 ≈ 21.4 m/s ≈ 77.1 km/h.

Q20. A mobile channel has a maximum Doppler shift of 200 Hz. Using T = 0.423 / f(d), estimate the channel's coherence time.

Ans: T = 0.423 / 200 ≈ 2.1 ms.

Q21. A GSM operator has 25 MHz in each direction, uses 200 kHz carriers and leaves one carrier as a guard. Each carrier carries 8 time slots. How many traffic channels are available?

Ans: 25 MHz / 200 kHz = 125 carriers; minus 1 guard = 124; 124 × 8 = 992 traffic channels.

Q22. A GSM burst period lasts 576.9 µs and the gross channel bit rate is 270.833 kbps. How many bits fit in one burst period?

Ans: Bits = 270.833 kbps × 576.9 µs ≈ 156.25 bits.

Q23. An IS-95 CDMA link spreads a 9.6 kbps voice stream at a chip rate of 1.2288 Mcps. What is the processing gain in dB?

Ans: Gp = 1.2288 Mcps / 9.6 kbps = 128; 10 log 128 ≈ 21.07 dB.

Q24. An FDMA system has 12.5 MHz of spectrum, 30 kHz channels and a 10 kHz guard band at each edge of the allocation. How many channels can it provide?

Ans: Usable spectrum = 12.5 MHz − 2 × 10 kHz = 12.48 MHz; channels = 12480 / 30 = 416.

Q25. An LTE carrier has 1200 usable subcarriers spaced 15 kHz apart, grouped 12 per resource block. How many resource blocks does the carrier offer?

Ans: Each RB = 12 subcarriers; 1200 / 12 = 100 resource blocks.

Q26. Handsets contend on a slotted-ALOHA random access channel with an offered load of G = 0.5 attempts per slot. What throughput does the channel achieve?

Ans: S = G·e^(−G) = 0.5 × e^(−0.5) = 0.5 × 0.6065 ≈ 0.303.

Q27. A location area has 25 cells, and the network must page 2000 incoming calls per hour in every cell of that area. How many paging messages are sent per hour?

Ans: 25 cells × 2000 = 50,000 paging messages per hour.

Q28. A BSC controls 40 BTSs. Each BTS has 3 sectors, each sector has 2 TRXs, and each TRX carries 8 time slots. How many time slots does the BSC manage?

Ans: 40 × 3 × 2 × 8 = 1920 time slots.

Q29. One MSC can serve 500,000 subscribers. How many MSCs does a city with 2.4 million subscribers need?

Ans: 2,400,000 / 500,000 = 4.8, so 5 MSCs are needed.

Q30. An Abis link runs on an E1 line of 2.048 Mbps, divided into 16 kbps sub-channels. How many sub-channels does the link carry?

Ans: 2048 kbps / 16 kbps = 128 sub-channels.

Q31. An HLR stores 1.5 KB of profile data per subscriber. How much storage is needed for 8 million subscribers?

Ans: 1.5 KB × 8,000,000 = 12,000,000 KB ≈ 12 GB.

Q32. A VLR serves 60,000 visiting subscribers, and each performs 4 location updates per hour. How many location updates per second must the VLR process?

Ans: Updates per hour = 60,000 × 4 = 240,000; per second = 240,000 / 3600 ≈ 66.7 updates/s.

Q33. A GSM full-rate speech codec outputs 13 kbps, and channel coding raises this to 22.8 kbps over the air. What percentage overhead does channel coding add?

Ans: Overhead = (22.8 − 13) / 13 = 0.754 ≈ 75.4%.

Q34. A 1400-byte voice payload is carried with a 20-byte IP header, an 8-byte UDP header and a 12-byte RTP header. What is the protocol efficiency?

Ans: Efficiency = 1400 / (1400 + 20 + 8 + 12) = 1400 / 1440 ≈ 97.2%.

Q35. An AMR codec produces a 244-bit speech frame every 20 ms, and each frame is sent with 40 bytes of IP/UDP/RTP headers. What bit rate does the stream use on the link?

Ans: Payload = 244 bits; headers = 40 × 8 = 320 bits; frame = 564 bits; rate = 564 / 20 ms = 28.2 kbps.

Q36. In the packet core, a 1000-byte user packet is tunnelled with an 8-byte GTP-U header, an 8-byte UDP header and a 20-byte IP header. What percentage of the transmitted bytes is tunnelling overhead?

Ans: Overhead = 8 + 8 + 20 = 36 bytes; total = 1036 bytes; 36 / 1036 ≈ 3.47%.

Q37. A GSM LAPDm signalling frame is 23 bytes long, of which 3 bytes are header. What is the frame's payload efficiency?

Ans: Payload = 23 − 3 = 20 bytes; efficiency = 20 / 23 ≈ 87%.

Q38. A PLMN uses a 15-digit IMSI with a 3-digit MCC and a 2-digit MNC. How many digits are left for the MSIN, and how many subscribers can the operator number?

Ans: MSIN digits = 15 − 3 − 2 = 10 digits; subscribers = 10^{10} (10 billion).

Q39. Another operator in the same country is assigned a 3-digit MNC. How many unique subscriber identities can it issue?

Ans: MSIN digits = 15 − 3 − 3 = 9; identities = 10^{9} (1 billion).

Q40. A core-network packet gateway handles 40 Gbps. In the busy hour, each user averages 2 Mbps when active and is active 10% of the time. How many users can the gateway support?

Ans: Average per user = 2 Mbps × 0.1 = 0.2 Mbps; users = 40,000 Mbps / 0.2 = 200,000 users.

Q41. On a data session, one-way delays are 10 ms on the radio link, 5 ms on backhaul, 3 ms in the core and 22 ms across the internet. What is the round-trip time?

Ans: One-way delay = 10 + 5 + 3 + 22 = 40 ms; RTT = 2 × 40 = 80 ms.

Q42. A network starts handoff when the received signal falls to -95 dBm, and the minimum usable signal for voice is -100 dBm. What is the handoff margin?

Ans: Handoff margin = −95 − (−100) = 5 dB.

Q43. A vehicle travels at 72 km/h along a highway lined with cells 2 km across. How many handoffs does it make in one hour?

Ans: Distance = 72 km/h × 1 h = 72 km; handoffs = 72 / 2 = 36.

Q44. A handoff margin of 6 dB is set. The signal fades by 0.1 dB per metre near the cell edge, and the user moves at 15 m/s. How much time does the network have to complete the handoff?

Ans: Distance for 6 dB fade = 6 / 0.1 = 60 m; time = 60 / 15 = 4 s.

Q45. A serving cell is received at -85 dBm and a neighbour at -82 dBm, with a 4 dB handoff hysteresis. Will handoff be triggered? If not, how much stronger must the neighbour become?

Ans: The neighbour is only 3 dB stronger (−82 vs −85), which is less than the 4 dB hysteresis, so handoff is NOT triggered. The neighbour must be more than 4 dB stronger, i.e. above −81 dBm (at least 1 dB more).

Q46. One million inbound roamers each make 2 location updates per day. Each update exchanges 5 signalling messages of 200 bytes. How much roaming signalling traffic is generated per day?

Ans: Messages = 1,000,000 × 2 × 5 = 10,000,000; traffic = 10^7 × 200 B = 2×10^9 bytes ≈ 2 GB per day.

Q47. A user crosses 12 cell boundaries per hour, and 20% of those crossings are also location-area boundaries. How many location updates does the user trigger per hour?

Ans: 12 × 0.20 = 2.4 location updates per hour.

Q48. A cell serves 2000 subscribers. Each makes 3 calls in the busy hour, and calls last 2 minutes on average. What traffic, in Erlangs, does the cell carry?

Ans: Traffic = (2000 × 3 × 2 min) / 60 = 200 Erlangs.

Q49. Network statistics show 900 dropped calls out of 45,000 established calls in a day. What is the call drop rate?

Ans: 900 / 45,000 = 0.02 = 2%.

Q50. A cell with 24 channels is offered 18 Erlangs at a 2% grade of service. How much traffic is actually carried, and what is the channel utilisation?

Ans: Carried traffic = 18 × (1 − 0.02) = 17.64 Erlangs; utilisation = 17.64 / 24 ≈ 73.5%.